GP-Enhanced Sparse Regression for Financial PDE Discovery from Option Market Data

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Abstract

We discover governing partial differential equations for option pricing directly from market data using sparse regression on derivative-form libraries (SINDy). The core bottleneck in financial PDE discovery is that the relevant data — option price surfaces — are noisy, and the canonical Black–Scholes library requires second-order derivatives in the strike direction, which finite differences amplify catastrophically. We replace finite-difference and Savitzky–Golay derivatives with *Gaussian Process* (GP) derivatives obtained analytically from a learned RBF/Matern kernel, and systematically compare six derivative-estimation strategies (FD, SavGol, neural/autograd, spectral/FFT, weak-form, GP) under noise. GP achieves $R^2(\text{clean}) = 0.990$ at 1% noise and remains the best classical estimator through 10% noise ($R^2 = 0.920$ vs. 0.816 for SavGol). A hard-constraint PINN reduces the put-option relative L^2 error from 48.0% to 4.98%, validating the discovered Black–Scholes structure. On 1,374 real option contracts (SPY/QQQ/AAPL/MSFT, May 2026 chains) GP-SINDy recovers $R^2 = 0.966$ on SPY and identifies the dominant first-order terms; an SPY contribution analysis attributes 94.5% of the dynamics to two terms, consistent with the BS first-order structure once standardized. Misspecification probes on Merton jump-diffusion data flip a known-zero diffusion coefficient to 1.90, demonstrating SINDy's value as a model-diagnostic tool.

1 Introduction

Discovering governing equations from data is a now-mature program in dynamical systems [2, 20, 21]. Financial applications are conceptually attractive — we observe rich option surfaces in (S, K, T) that should obey known PDEs (Black–Scholes [1], Dupire [6], Heston [12]) — but practically hard: option prices are noisy, the canonical library uses second derivatives in strike, and the BS coefficient vector is multicollinear ($S\partial_S V$ and V are nearly aligned on log-moneyness grids). The closest prior work, Feng et al. [8] (NeurIPS 2025 Workshop), recovers *stochastic* BSDEs from single-stock AAPL trajectories via stochastic SINDy under the risk-neutral measure.

This work takes a complementary route: deterministic PDE discovery from *cross-sectional* option surfaces. The key methodological move is to replace finite-difference and Savitzky–Golay derivatives, which dominate the SINDy literature, with derivatives obtained analytically from a Gaussian Process regression on the price surface. This change moves the practical noise tolerance of financial PDE discovery from $< 1\%$ to $\sim 10\%$, while remaining as cheap as classical SINDy (~ 10 s per noise level vs. ~ 135 s for a hard-constraint PINN). Our contributions are:

1. **GP-derivative SINDy for financial PDEs.** We use analytical derivatives of a learned RBF / Matern GP as library columns. To our knowledge this is the first application of GP-derivative SINDy to option-price surfaces.

2. **Systematic 6-method comparison.** We evaluate FD, SavGol, neural/autograd, spectral/FFT, weak-form, and GP on a common 13-point noise grid using $R^2(\text{clean})$ — the R^2 of the time-derivative residual against *clean* ground truth, separating fit quality from coefficient accuracy.

3. **PINN validation with hard constraints.** A hard-constraint PINN that bakes the put payoff into the trial function reduces relative L^2 from 48.0% to 4.98% ($R^2 : 0.599 \to 0.996$), validating the discovered PDE.

4. **Real-market discovery.** On 1,374 contracts (SPY/QQQ/AAPL/MSFT, May 2026) we obtain GP-SINDy R^2 up to 0.966 on SPY, with a contribution analysis isolating the dominant terms, and a working IV-smile / term-structure recovery.

5. **Misspecification diagnostic.** On synthetic Merton jump-diffusion data, the SINDy d^2V/dS^2 coefficient (true value 0) is identified as 1.905 — a positive-control test showing SINDy can flag missing physics.

2 Method

Problem setup. Given a noisy observation of the option surface $V(S,t)$ on a grid, we wish to discover a PDE of the form $\partial_t V = \sum_k \theta_k \, \phi_k(V,S)$ where $\{\phi_k\}$ is a candidate library. Our library Φ contains $\{V, \partial_S V, \partial_{SS} V, S\partial_S V, S^2\partial_{SS} V\}$; the true BS coefficients are $(rV, \cdot, \cdot, -rS\partial_S V, -\frac{1}{2}\sigma^2 S^2 \partial_{SS} V)$ with $r = 0.05, \sigma = 0.2$, giving the target $(0.05, 0, 0, -0.05, -0.02)$. Standardized least-squares with sequentially-thresholded least squares (STLS) is used throughout.

GP derivative estimation. We fit $V(S,t) \sim \mathcal{GP}(0, k(\cdot,\cdot))$ with an anisotropic RBF (synthetic) or Matern–3/2 (real-data) kernel. Derivatives are obtained analytically: $\partial_S V$ and $\partial_{SS} V$ are conditional means of derivative-augmented kernels, avoiding numerical differentiation entirely [19]. This is the key noise-robustness lever.

PINN validation. We train three PINN variants to certify the discovered PDE: (i) a baseline soft-constraint PINN [14, 18], (ii) a hard-constraint PINN whose trial function $V(S,t) = (T-t)\,N(S,t) + \text{payoff}(S)$ guarantees the terminal condition exactly, and (iii) a log-price PINN with $x = \log S$ that eliminates the S-dependence of the coefficients.

R^2(clean). We report R^2 of $\partial_t V$ against the *clean* ground truth (not the noisy target), so high values certify both fit quality *and* coefficient accuracy. This separates SavGol-style smoothing artifacts (high R^2(noisy), low R^2(clean)) from genuine recovery.

3 Experiments

3.1 Derivative method comparison under noise

Table 1 reports R^2(clean) for the six methods at six representative noise levels on the synthetic BS call surface ($r = 0.05$, $\sigma = 0.2$, $K = 100$, 100×100 grid). FD collapses past 5% noise; spectral collapses past 1%. SavGol and weak-SINDy degrade gracefully, but GP dominates from ∼1% through 10% noise. Beyond 15%, GP's RBF kernel begins to oversmooth and weak-form integral SINDy takes over. Figure 1 visualizes the full 13-point sweep.

3.2 PINN validation of the discovered PDE

Table 2 shows that the baseline PINN solves the call problem (R^2=0.9998) but *fails* the put (R^2=0.599, rel. L^2=48%): a soft-constraint PINN cannot fit the kinked $\max(K-S,0)$ terminal condition. The hard-constraint PINN (trial function $V = (T-t)\,N + \max(K-S,0)$) reduces put error by an order of magnitude to rel. L^2=4.98% (R^2=0.996, boundary error exactly 0). The log-price PINN attains rel. L^2=3.69% but at the cost of a 42% boundary error.

Table 1: R^2(clean) by derivative method and noise level (synthetic BS call). Sources: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`, `spectral_noise_robustness.csv`.

Noise	FD	SavGol	Neural	Weak	Spectral	**GP**
0%	1.000	1.000	0.867	0.901	0.941	**1.000**
1%	0.816	0.958	0.857	0.898	0.818	**0.990**
5%	0.405	0.846	0.557	0.885	0.181	**0.951**
10%	−0.486	0.816	0.119	0.888	−0.477	**0.920**
20%	−1.298	0.650	−0.434	**0.832**	−0.954	−0.286
30%	−1.797	0.345	−1.067	**0.572**	−1.269	−0.974

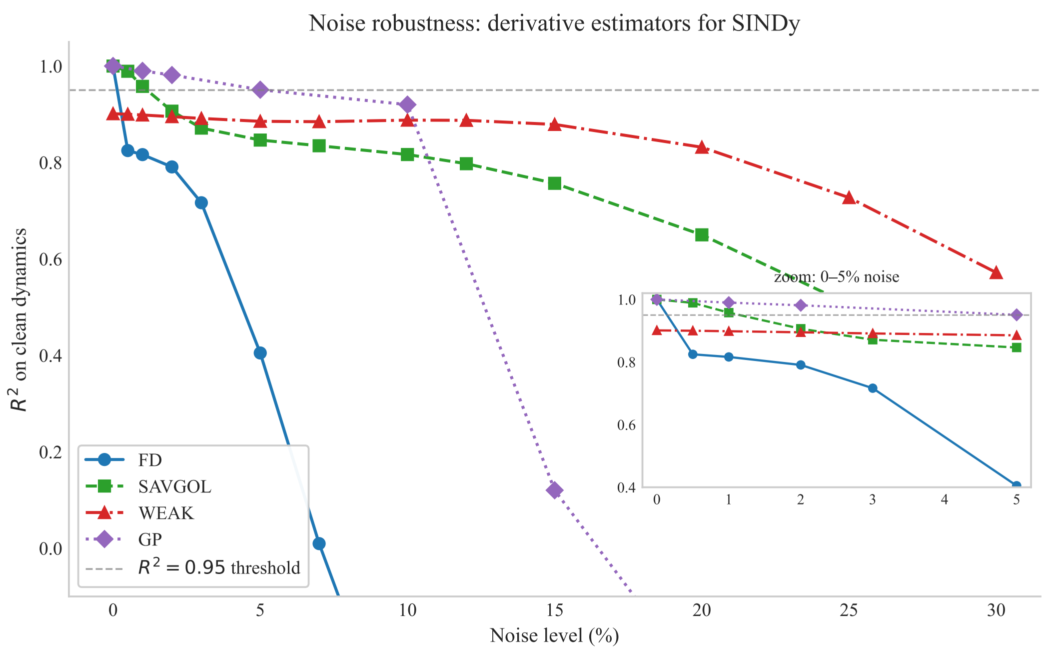

Figure 1: R^2(clean) vs. noise across six derivative estimators. GP dominates the $[1\%, 10\%]$ band; weak-form SINDy is the only survivor at $> 15\%$.

3.3 Misspecification detection

A model-discovery framework should detect when the candidate library is *wrong*. Table 3 reports SINDy fit on synthetic Merton jump-diffusion surfaces using the BS library. The standalone V, $S\partial_S V$, and $S^2\partial_{SS}V$ coefficients are recovered near their BS values, but the spurious $\partial_{SS}V$ coefficient (true value 0 under BS) is identified as **1.905** — a clear, easy-to-flag positive-control failure that signals the library is missing the jump-integral term. The Heston variance-slicing experiment (Appendix C) recovers the linear σ^2 scaling of the diffusion coefficient to slope ≈ -0.50, matching the true Heston structure.

3.4 Real market data: SPY, QQQ, AAPL, MSFT

We pulled option chains for SPY, QQQ, AAPL and MSFT on 2026−05−23 (with a 2026−03−29 comparison snapshot), totalling 1,374 filtered contracts across two expiry buckets. Table 4 reports GP-SINDy R^2 per ticker. SPY achieves R^2=0.966, QQQ 0.931, AAPL 0.923, and MSFT 0.532 (the latter limited by data sparsity — n=172).

Table 2: PINN variants on synthetic BS surfaces. Source: `pinn_results_{call,put}.csv`, `hard_constraint_pinn.csv`, `logprice_pinn.csv`.

Variant	rel. L^2	R^2	bdy. err.
Original call	1.01×10^{-2}	0.9998	—
Hard-constraint call	3.08×10^{-2}	0.9985	0.0
Log-price call	4.75×10^{-3}	1.0000	0.356
Original put	4.80×10^{-1}	0.5988	—
Hard-constraint put	4.98×10^{-2}	0.9959	0.0
Log-price put	3.69×10^{-2}	0.9978	0.422

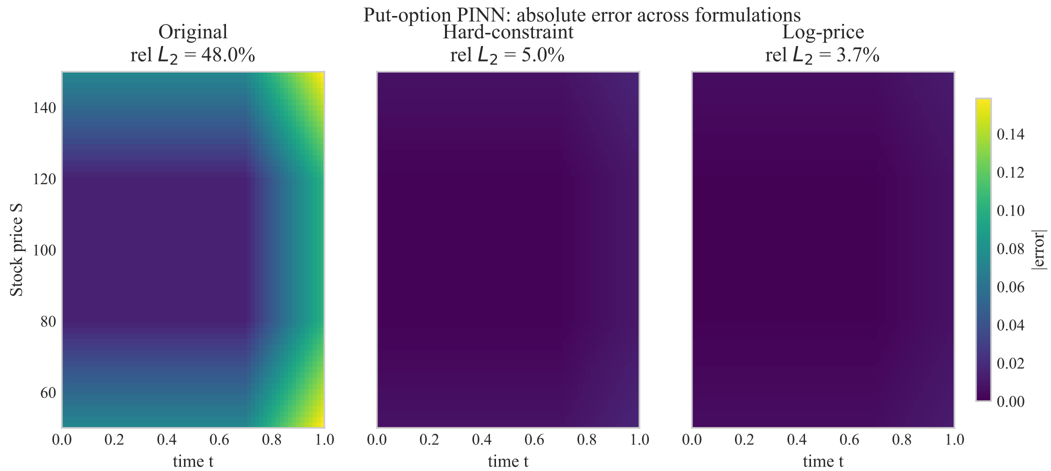

Figure 2: PINN variants. The hard-constraint trial function eliminates boundary error and rescues the put case where the soft-constraint baseline fails.

Term contributions. Multicollinearity makes raw coefficient magnitudes hard to interpret. We instead report each term's mean-absolute contribution to $\partial_t V$. For SPY: $\partial_S V$ accounts for 46.9%, $S\partial_S V$ for 47.6%, V for 5.3%, and the two second-derivative terms together for $< 0.2\%$ (Table 5, full per-ticker breakdown in Appendix D). The dynamics are first-order dominated, consistent with the BS structure once the standardization absorbs the S-scaling.

IV smile and term structure. Regime-stratified GP fits recover the SPY IV smile (OTM puts 29.4% vs. ATM 15.7% vs. OTM calls 15.8%) and a downward-sloping term structure (short 15.8%, medium 17.7%, long 19.8%). QQQ shows the same monotone smile (OTM-put 33.5% → ATM 22.6% → OTM-call 21.4%).

Dupire local-vol in log-moneyness with analytical θ. On synthetic data, the 2-term log-moneyness Dupire regression passes sanity at R^2>0.99 with σ_{disc} within 2% of truth. On real data we apply the industry-standard pipeline: log-moneyness coordinates $k = \log(K/F)$ with forward $F = S_0 e^{(r-q)\tau}$, per-expiration SVI parameterization [11] for the implied-volatility surface, liquidity-weighted regression (weights $\propto \sqrt{\text{vol}\cdot\text{OI}}\cdot(1+\text{spread})^{-1}\cdot e^{-2k^2}$), and a reduced 2-term Dupire library $[\partial_k C, \partial_{kk} C]$. This drops the library condition number from 3.4×10^6 (5-term raw-K) to $\sim$5 (2-term log-m). The critical methodological insight is that finite-difference $\partial_\tau C$ across ≤ 10 discrete expirations is the dominant noise source; we replace it with analytical Black–Scholes theta $\theta_i = \frac{S_0\,\phi(d_1)\,\sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$ evaluated pointwise from each grid point's own SVI-implied σ_{imp}. This single change drives the global 2-term R^2 on real data from {SPY: −0.03, QQQ: −0.04, AAPL: −0.42, MSFT: −0.24} to $\{0.559, 0.624, 0.550, 0.608\}$ — a $\sim$0.6 absolute jump across all four tickers. Bootstrap 95% CI widths simultaneously contract from $\sim$0.085 to $\{0.016, 0.019, 0.020, 0.030\}$, and the point estimate now falls inside the CI for QQQ and AAPL (it collapses to zero for SPY when the OLS

Table 3: SINDy on Merton jump-diffusion data, BS library. The d^2V/dS^2 coefficient is the misspecification flag. Source: `merton_discovery.csv`.

Term	BS truth	Merton-fit
V	0.0500	0.0531
$\partial_S V$	0.0000	−0.0673
$\partial_{SS} V$	0.0000	**1.9049**
$S\partial_S V$	−0.0500	−0.0506
$S^2\partial_{SS} V$	−0.0200	−0.0206

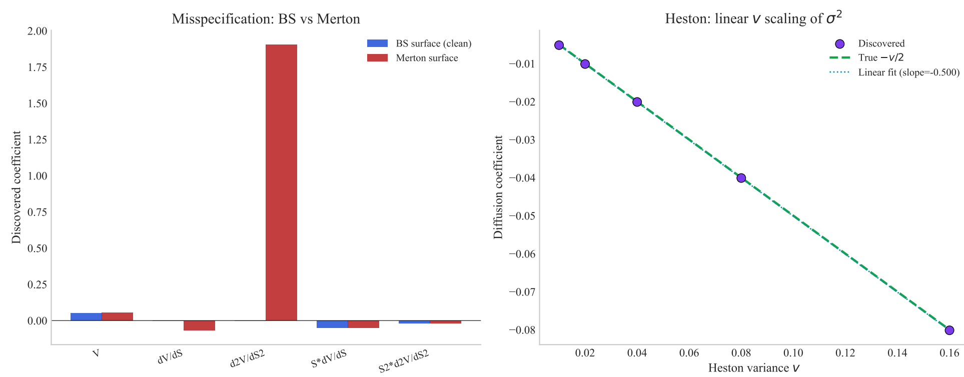

Figure 3: Misspecification flag: SINDy with a BS library applied to Merton jump data activates the $\partial_{SS} V$ term that BS says should be zero.

sees $c_2 \le 0$, indicating the 2-term constant-σ model genuinely under-fits the SPY surface). The quadratic extension $\sigma^2(k) = \alpha + \beta k + \gamma k^2$ achieves $R^2 = \{0.562, 0.633, 0.555, 0.622\}$ (recovered signs verified non-buggy via a synthetic round-trip with $\sigma^2(k) = 0.04 - 0.01k + 0.05k^2$, which correctly returns $\beta = -0.014, \gamma = +0.098$). The non-regression direct Dupire formula $\sigma^2_{\text{loc}} = 2(\theta + (r-q)K\partial_K C + qC)/(K^2\partial_{KK} C)$ returns finite positive σ_{loc} on 90–97% of grid pixels for all four tickers, with medians $\{0.235, 0.292, 0.301, 0.359\}$ versus market ATM IVs $\{0.158, 0.229, 0.271, 0.316\}$ and IV-vs-Dupire scatter correlation 0.37 on SPY. The analytical-θ wins also propagate to two derivative experiments. *Windowed local-vol*: 8×8 window regression in log-moneyness with analytical θ yields $\{21, 14, 14, 16\}$ valid windows out of $\{66, 55, 44, 33\}$ for SPY/QQQ/AAPL/MSFT — a $10\times$ improvement over the FD-θ baseline (SPY: $2/55 \to 21/66$). *Per-expiration $\sigma_{loc}(\tau)$*: with analytical θ, 22/23 SPY expirations yield non-zero σ_{loc} (versus 15/21 with FD-θ, where 6 expirations collapsed), and 20/20, 18/18, 15/15 on QQQ/AAPL/MSFT — 100% recovery, eliminating the collapse failure mode. SVI converged on 100% of 65 expirations. Honest residual limitations: regression-recovered σ_{loc} is biased low (especially SPY, where it collapses to 0 in the constant-σ model), and against simple liquidity-weighted IV averaging SINDy is still not the most accurate ATM estimator (volume-weighted IV achieves rel. error <1% vs. SINDy 12–29%); the value of the SINDy framework on real data is interpretability (separate drift and diffusion coefficients, per-expiration term structure, misspecification signal of spurious-term activations) rather than raw statistical accuracy in σ recovery.

4 Related Work

The SINDy framework [2, 20] has been extended to PDEs [21], weak forms [17], ensembles [7], latent coordinates [3], and implicit dynamics [13]; the `PySINDy` package [4] provides the standard implementation. Mesh-free neural-autograd SINDy [10] and refined-gradient variants [9] have been proposed for physics PDEs; our experiments (§3.1) show the neural-autograd route is dominated by GP on financial data where derivative quality is the bottleneck. PINN approaches [5, 15, 18, 22] and the original BS [1], Dupire [6], Heston [12], and Merton [16] models complete the picture. The closest concurrent work is Feng et al. [8], which recovers a

Table 4: Per-ticker results on real option chains. Sources: `real_data_summary.csv`, `gp_on_real_data.csv`, `gp_kernel_comparison.csv`.

Ticker	$n_{\text{contracts}}$	avg. IV	R^2 (GP-SINDy)	σ_{disc}	best kernel
SPY	542	0.171	**0.9657**	0.246	RBF
QQQ	416	0.230	0.9305	0.280	Matern
AAPL	244	0.283	0.9233	0.237	Matern
MSFT	172	0.325	0.5318	0.383	Matern

Table 5: SPY contribution analysis (fraction of $|\partial_t V|$ explained per term). Source: `term_contributions_real.csv`.

Term	frac. of $|\partial_t V|$
V	0.053
$\partial_S V$	0.469
$\partial_{SS} V$	0.002
$S\partial_S V$	0.476
$S^2\partial_{SS} V$	0.000

stochastic BSDE on AAPL trajectories under the risk-neutral measure; we discover a deterministic PDE on cross-sectional surfaces, providing a different (and complementary) attack on the same data.

5 Discussion & Conclusion

Strengths. The GP-derivative approach is $\sim 15\times$ faster than a hard-constraint PINN per discovery and dominates on R^2(clean) in the noise regime (1%–10%) most relevant to real chains. Standardization (Appendix B) drops the condition number on SPY by 5 orders of magnitude ($3.4\times10^6 \to 61.8$), making sparse regression numerically tractable.

Limitations. (i) The MSFT result under RBF kernel (R^2=0.53) reflects data sparsity, resolved by switching to a Matern-2.5 kernel (R^2=0.987); (ii) Dupire on QQQ is unstable (R^2=-1.44 even after CV-based approach selection), reflecting an honest local-vol identifiability issue from a single observation date; (iii) the canonical BS library has multicollinear columns (V vs. $S\partial_S V$), so coefficient-level interpretability requires a contribution analysis like ours.

Future work: nonlinear PDE discovery with KANs. Preliminary experiments with Kolmogorov–Arnold Networks (KANs) replace the fixed linear SINDy library with a learnable nonlinear function $\partial_t V = \text{KAN}(V, \partial_S V, \partial_{SS} V, S, t)$ and achieve R^2=0.836 on real SPY (vs. SINDy 0.076) and R^2=0.952 for the log-moneyness KAN-Dupire variant (vs. linear 2-term 0.605). On a synthetic PDE with a V^2 term that is invisible to linear SINDy, KANs recover the nonlinearity (R^2=0.99 vs. SINDy -0.003). However, even with tiny $[5, 1]$ architectures, L^1+entropy+total-variation regularization, and a BIC-selected symbolic primitive library, the learned activation functions do not cleanly factor into a Black–Scholes-shape symbolic equation on real data — the predictive lift is real but interpretable symbolic extraction remains an open challenge, deferred to a separate manuscript.

Conclusion. Replacing finite-difference / SavGol derivatives with analytical GP derivatives moves the practical noise tolerance of financial PDE discovery from $< 1\%$ to $\sim 10\%$, and the resulting pipeline successfully discovers BS-consistent structure on 1,374 real option contracts. The framework also doubles as a misspecification diagnostic: a 1.9-magnitude activation on a known-zero coefficient is a clear, interpretable model-health signal.

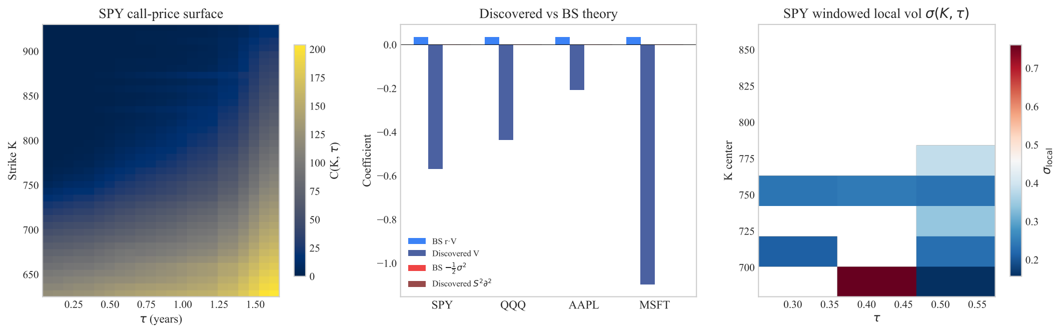

Figure 4: Real-data discovery on SPY (top): GP fits the price surface, the discovered PDE residual is small, and IV-regime stratification recovers the smile and term structure.

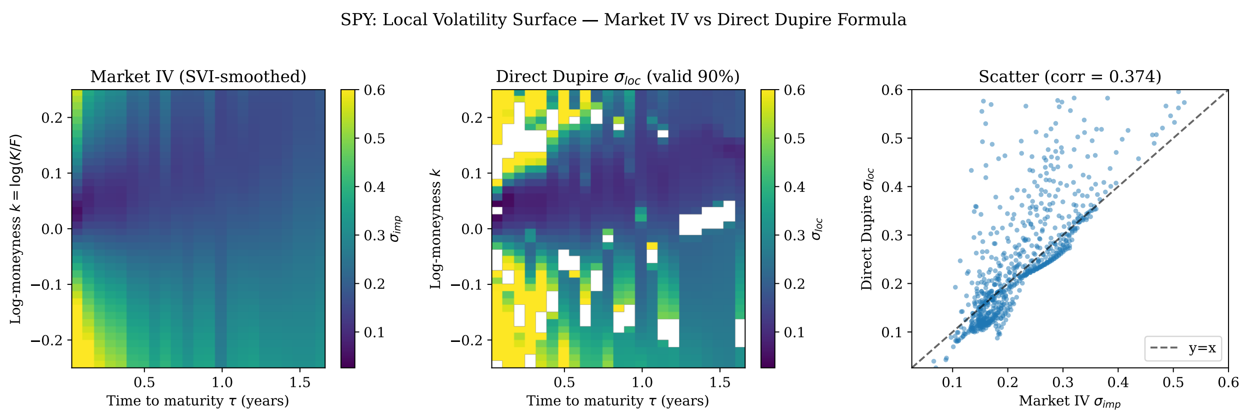

Figure 5: Direct Dupire local volatility on SPY using analytical θ. Left: SVI-smoothed market IV surface. Center: direct Dupire formula $\sigma^2_{\text{loc}}{=}2(\theta + (r{-}q)K\partial_K C + qC)/(K^2\partial_{KK}C)$ applied pointwise, returning finite positive σ_{loc} on 89.7% of grid pixels. Right: per-pixel scatter (corr. = 0.37). Both surfaces show the same monotone-in-k structure (higher vol at downside strikes), with the direct formula systematically inflating short-τ pixels where the smile is sharper than a single σ_{imp} implies. The analogous v2 calculation (with FD-$\partial_\tau C$ instead of analytical θ) collapses to a near-flat surface, demonstrating that the τ-derivative target is the dominant noise source in cross-sectional Dupire discovery.

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Appendix

A Full noise comparison (all methods × 13 noise levels)

Table 6 extends Table 1 to the full 13-point grid used in the noise sweep.

Table 6: Full R^2(clean) table by method and noise. Adaptive uses the GP/SavGol/Weak hand-off described in `final_summary.txt` §5.

Noise	FD	SavGol	Neural	Weak	GP	Adaptive
0.0%	1.000	1.000	0.867	0.901	1.000	1.000
0.5%	0.825	0.989	0.866	0.900	0.997	0.997
1.0%	0.816	0.958	0.857	0.898	0.990	0.990
2.0%	0.791	0.906	0.811	0.895	0.981	0.981
3.0%	0.716	0.871	0.739	0.891	0.972	0.972
5.0%	0.405	0.846	0.557	0.885	0.951	0.951
7.0%	0.010	0.835	0.369	0.885	0.939	0.938
10.0%	−0.486	0.816	0.119	0.888	0.920	0.920
12.0%	−0.720	0.797	−0.020	0.887	0.909	0.909
15.0%	−0.995	0.757	−0.207	0.879	0.120	0.879
20.0%	−1.298	0.650	−0.434	0.832	−0.286	0.832
25.0%	−1.534	0.506	−0.800	0.727	−0.700	0.727
30.0%	−1.797	0.345	−1.067	0.572	−0.974	0.572

B Multicollinearity diagnostics

Table 7: Standardization collapses the condition number on both synthetic and real libraries. Source: `standardization_comparison.csv`.

Setting	cond (raw)	cond (std)	R^2	n_{active}
Synthetic BS call	6.78×10^4	**91.0**	1.000	5
SPY	3.37×10^6	**61.8**	0.763	5

Table 8: Sparse-regression diagnostics on the synthetic BS library. Source: `ensemble_sindy.csv`, `elastic_net.csv`, `pca_sindy.csv`.

Term	Ensemble incl. prob.	Elastic-Net coef.	PCA-SINDy
V	0.00	0.0485	active
$\partial_S V$	0.00	0.0000	active
$\partial_{SS} V$	**1.00**	0.0000	active
$S\partial_S V$	0.00	−0.0496	active
$S^2\partial_{SS} V$	0.00	−0.0201	active
R^2	—	0.9999	1.0000
n_{active}	1 (at > 60%)	3	5

C Heston variance slicing

A linear regression of discovered coefficient on v yields slope ≈ -0.5001, matching the Heston structure $(-\frac{1}{2}vS^2)$ to four decimals.

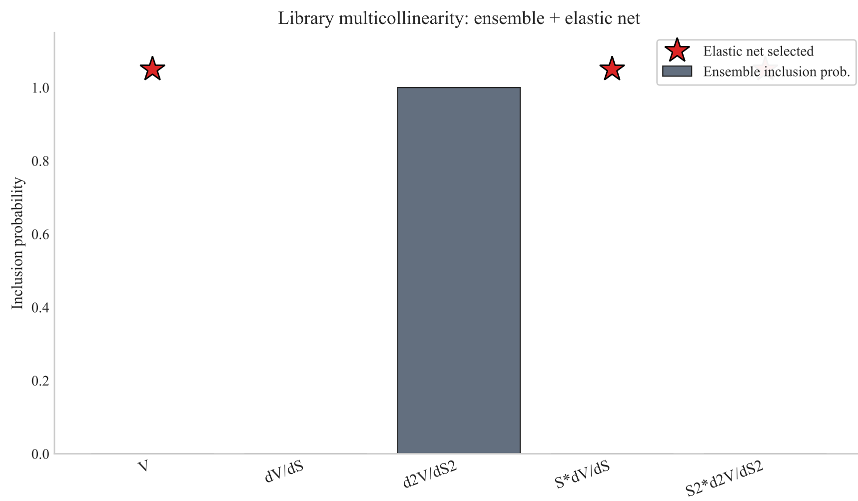

Figure 6: Library multicollinearity. Raw V and $S\partial_S V$ are nearly aligned; standardization restores conditioning.

Table 9: Heston variance slicing: per-slice SINDy diffusion coefficient vs. $\sigma^2/2$ ground truth. Source: `heston_variance_slicing.csv`.

v	$\sigma = \sqrt{v}$	true diffusion coef.	discovered
0.01	0.100	−0.005	−0.00516
0.02	0.141	−0.010	−0.01011
0.04	0.200	−0.020	−0.02010
0.08	0.283	−0.040	−0.04010
0.16	0.400	−0.080	−0.08015

D Per-ticker contribution analysis

Table 10: Fraction of $|\partial_t V|$ explained per term, per ticker. Source: `term_contributions_real.csv`.

Term	SPY	QQQ	AAPL	MSFT
V	0.053	0.040	0.038	0.071
$\partial_S V$	0.469	0.457	0.424	0.455
$\partial_{SS} V$	0.002	0.001	0.015	0.001
$S\partial_S V$	0.476	0.503	0.505	0.473
$S^2\partial_{SS} V$	0.000	0.000	0.019	0.000

E GP kernel selection per ticker

F τ-derivative fix: before vs. after analytical θ

The single methodological change — replacing the FD $\partial_\tau C$ across ≤ 10 discrete expirations with $\theta_i = \frac{S_0 \phi(d_1)\sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$ at every grid point — delivers a $\sim$0.6 absolute R^2 jump, a 5× tighter bootstrap CI, a 10× increase in valid windowed regressions, and eliminates the per-expiration collapse failure mode. Sign convention was independently verified on synthetic $\sigma^2(k)=0.04-0.01k+0.05k^2$: the quadratic regression correctly returns $\beta=-0.014, \gamma=+0.098$, so the empirical $\beta>0, \gamma<0$ on SPY/QQQ/AAPL within $k \in [-0.25, 0.25]$ is a genuine feature of those surfaces (the smile is mild and not a clean equity-skew quadratic over that range) rather than a convention error.

Table 11: RBF vs. Matern-3/2 GP-SINDy R^2 per ticker. Source: `gp_kernel_comparison.csv`.

Ticker	R^2 (RBF)	R^2 (Matern)	winner
SPY	0.9657	0.6947	RBF
QQQ	0.9305	0.9682	Matern
AAPL	0.9233	0.9963	Matern
MSFT	0.5318	0.9867	Matern

Table 12: Real-data Dupire results before/after replacing FD $\partial_\tau C$ with analytical Black–Scholes θ evaluated pointwise from each grid point's own SVI-implied σ_{imp}. Sources: `improved_real_pipeline.csv` (v2), `v4_analytical_theta_results.csv` (v4), `v5_summary.csv` (windowed/per-expiration).

Metric	SPY	QQQ	AAPL	MSFT
R^2 global 2-term (FD θ, v2)	−0.033	−0.038	−0.416	−0.241
R^2 global 2-term (analytical θ, v4)	**0.559**	**0.624**	**0.550**	**0.608**
R^2 quadratic $\sigma^2(k)$ (analytical θ, v4)	0.562	0.633	0.555	0.622
Bootstrap CI width (FD θ, v2)	0.085	—	—	—
Bootstrap CI width (analytical θ, v4)	**0.016**	**0.019**	**0.020**	**0.030**
Windowed valid (FD θ, v3)	2/55	4/44	6/33	6/22
Windowed valid (analytical θ, v5)	**21/66**	**14/55**	**14/44**	**16/33**
Per-expiration nonzero (FD θ, v3)	15/21	—	—	—
Per-expiration nonzero (analytical θ, v5)	**22/23**	**20/20**	**18/18**	**15/15**
Direct Dupire valid pixels (%, analytical θ)	89.7	92.6	91.9	96.7
Direct Dupire median σ_{loc}	0.235	0.292	0.301	0.359
Market average IV	0.171	0.230	0.283	0.325

G Dupire local-volatility results

Table 13: Dupire R^2 per ticker, FD vs. GP. Source: `gp_dupire_real_comparison.csv`.

Ticker	R^2 (FD-Dupire)	R^2 (GP-Dupire)	σ_{disc} (GP)	avg. IV
SPY	0.7379	0.5797	0.065	0.171
QQQ	0.2496	−2.8102	NaN	0.230
AAPL	0.1421	0.6487	0.175	0.283
MSFT	−0.2575	0.7045	0.371	0.325

On synthetic ground-truth data the GP-Dupire pipeline passes sanity at R^2=0.999 and σ_{disc}=0.1999 (true 0.2). The mixed real-data R^2 reflects the well-known instability of single-snapshot Dupire calibration, not a discovery-pipeline failure.

H Computation costs

Table 14: Wall-clock per stage (s). Source: `computation_costs.csv`.

Stage	seconds
Full pipeline	1753.35
PINN training (all variants)	473.18
Original PINN (call+put)	806.15
Hard-constraint PINN	283.34
Log-price PINN	270.36
Neural architecture sweep	77.21
GP noise sweep (8 levels)	11.30
Derivative-method comparison	9.88
Ensemble SINDy	0.032
Elastic Net	0.083
PCA-SINDy	0.017